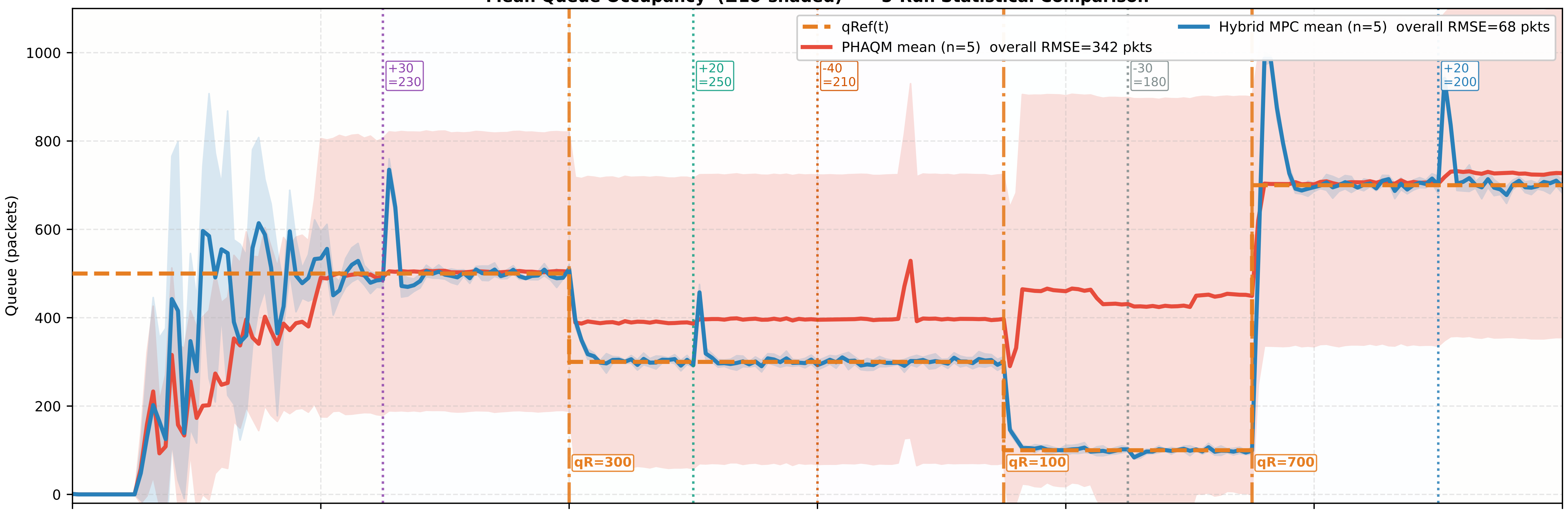
PHAQM vs Hybrid MPC+PID+RLS — Dynamic AQM Comparison

5 Independent Runs | 200 Heterogeneous Flows | 45 Mbps Bottleneck | 8 Dynamic Events

Mean Queue Occupancy (±1σ shaded) — 5-Run Statistical Comparison



Overall Improvement

5.1x

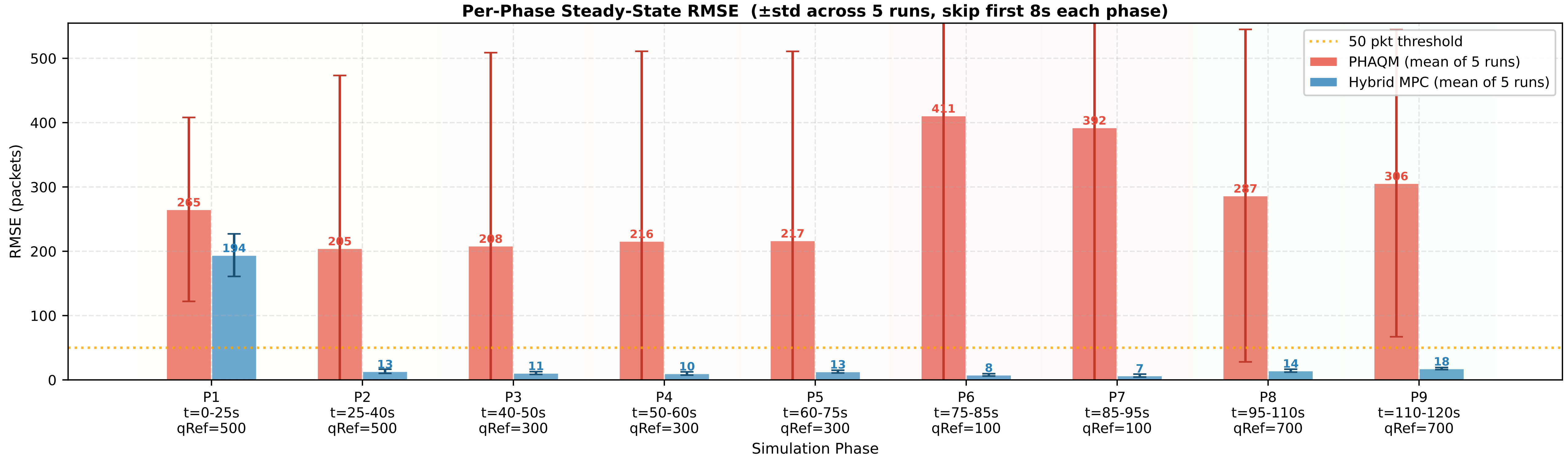
better RMSE

Hybrid MPC vs PHAQM

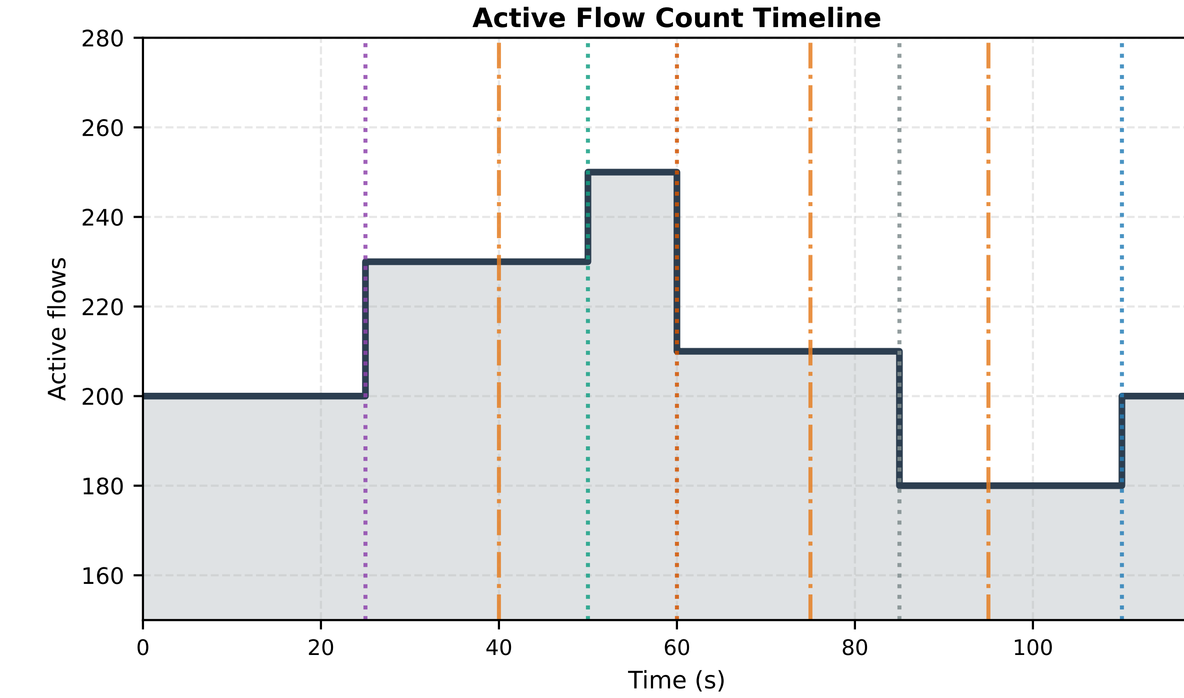
PHAQM: 342 pkts

Hybrid: 68 pkts

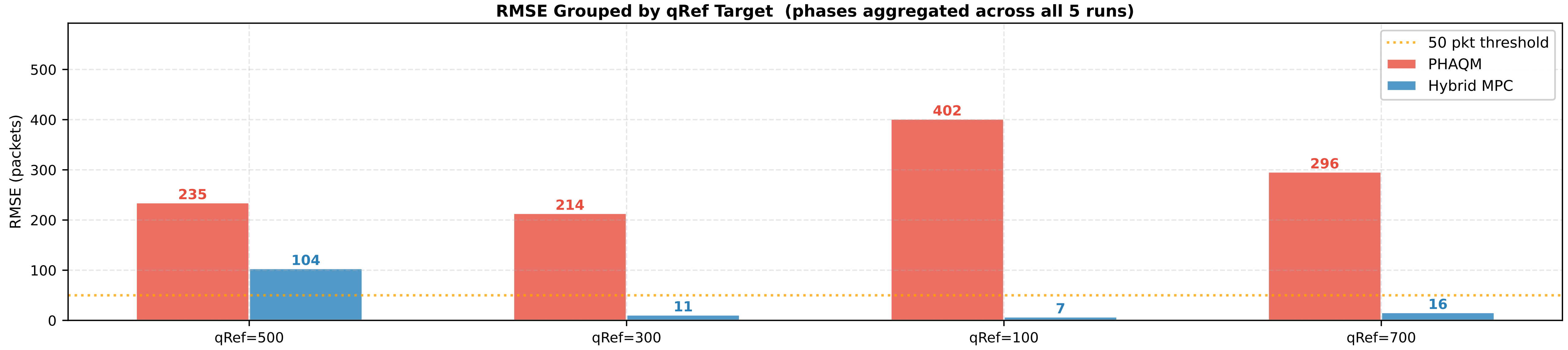
Per-Phase Steady-State RMSE (±std across 5 runs, skip first 8s each phase)



Active Flow Count Timeline



RMSE Grouped by qRef Target (phases aggregated across all 5 runs)



RMSE per qRef Level

qRef	PHAQM RMSE	MPC RMSE	Improv.
500	235	104	2.3x
300	214	11	19.0x
100	402	7	55.0x
700	296	16	18.5x
ALL	342	68	5.1x

Per-Phase ITU-T Compliance Summary (RMSE < 50 pkts threshold; post-8s settling window)

Phase	Period	qRef	PHAQM RMSE	MPC RMSE	Improv.	PHAQM ITU-T	MPC ITU-T
Phase 1	t=0-25s	500	265 pkts	194 pkts	1.4x	FAIL	FAIL
Phase 2	t=25-40s	500	205 pkts	13 pkts	15.3x	FAIL	PASS
Phase 3	t=40-50s	300	208 pkts	11 pkts	19.4x	FAIL	PASS
Phase 4	t=50-60s	300	216 pkts	10 pkts	21.6x	FAIL	PASS
Phase 5	t=60-75s	300	217 pkts	13 pkts	16.7x	FAIL	PASS
Phase 6	t=75-85s	100	411 pkts	8 pkts	52.1x	FAIL	PASS
Phase 7	t=85-95s	100	392 pkts	7 pkts	58.4x	FAIL	PASS
Phase 8	t=95-110s	700	287 pkts	14 pkts	19.9x	FAIL	PASS
Phase 9	t=110-120s	700	306 pkts	18 pkts	17.3x	FAIL	PASS